

Pouyan Asgharian

Robotics Engineer

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Website 

Summary

PhD Candidate in Robotics at Université de Sherbrooke specializing in automation, robot learning, human-robot interaction, and control. Dedicated to translating advanced AI-based pipelines into intuitive industrial solutions. Hands-on experience developing end-to-end automation pipelines, integrating diverse robotic arms, vision systems, and hardware-software interfaces.

Education

PhD Candidate in Robotics Université de Sherbrooke	2021 - present
MSc in Electrical Engineering University of Zanjan	2014 - 2016
BSc in Electrical Engineering Azarbaijan Shahid Madani University	2010 - 2014

Skills

Robotic Platforms	Mecademic Meca500 Kinova (Gen3, Gen3 lite, Link6) Universal Robots UR5 TIAGo PAL Robotics
Robotics Frameworks	ROS / ROS2 MoveIt Gazebo MuJoCo PyBullet Webots RoboDK
Programming	Python C MATLAB
ML & VLM Tools	PyTorch OpenCV TorchVision scikit-learn YOLO (v8, YOLOE) Optuna LangChain Ollama
Embedded Systems	Arduino Raspberry Pi ESP32 AVR
DevOps & Tools	Linux (Ubuntu) Git Docker Simulink Proteus FreeCAD
Soft Skills	Technical Communication Team Collaboration Adaptability

Professional Experience

Pfizer Developing an automated laboratory workflow using the Mecademic Meca500 robotic arm, with a focus on reducing task-specific hard coding. Designed a human-in-the-loop robotic system combining vision-language models (VLMs) and interactive imitation learning to enable intuitive teaching and adaptation of lab processes.	Apr. 2024 – Dec. 2025 Montreal, QC
National Research Council (NRC), Canada As a part-time volunteer researcher, contributed to electronic circuit design and robotic simulation for a collaborative cobotic drilling project.	Jan. 2024 – Feb. 2024 Montreal, QC
Acrylic Robotics Robotics engineering intern at an art-tech startup, contributing to the development and deployment of robotic systems for automated art production. Led hands-on integration and operation of Mecademic Meca500 and Kinova robotic arms, and supported live demos, exhibitions, and technical workshops.	Sept. 2022 – Dec. 2023 Montreal, QC

Awards & Grants

Recipient of Mitacs Business Strategy Internship (BSI) grant (×6)	2023-25
Scholarship recipient under the CoRoM program (CRSNG FONCER)	2023

Winner of 2 Minutes of Engineering competition - Université de Sherbrooke	2022
Awarded the Eureka Scholarship - Université de Sherbrooke	2021

Leadership & Engagement

Science Communication

- Robo Tik Podcast (2022–Present): Host of a Persian-language podcast about AI & Robotics.
- Sherbotics Podcast (2021–2022): Previous Host of an English-language robotics podcast in collaboration with CFAK radio.

Academic Service

- Judge at RoboHacks Hackathon - McGill University(2023)
- Member of the IEEE Robotics and Automation Society and the IEEE Computer Society
- Student board member of the HCI International Conference (2021–2022)
- Reviewer for peer-reviewed journals such as IEEE Robotics and Automation Letters (RA-L)

Teaching & Volunteer Mentorship

- Served as a Lecturer (Mooj Institute) and Teaching Assistant (U. of Zanjan) for 8+ courses, including Industrial Control, Electronics Lab, and Electrical Machinery.
- Mentor at Université de Sherbrooke (Fall 2023, Fall 2025)
- Mentor at the Women in Machine Learning (WiML) Mentorship Program (2023 – 2024)
- Active Mentor at the Mentspot website (2022–present)

Publications

- [1] Pouyan Asgharian, Adina M. Panchea, François Ferland, “A Review on the Use of Mobile Service Robots in Elderly Care”, Robotics, Vol. 11, No. 5, 2022.
- [2] Pouyan Asgharian, Zati Hakim Azizul, “Proposed Efficient Design for Unmanned Surface Vehicles”, arXiv preprint arXiv:2009.01284, 2020.

*Full publication list: pouyanasgharian.com/pub

Certifications

Québec Scientific Entrepreneurship (QcSE) Program	2025
Montreal Robotics Summer School (MRSS) - Mila, Montreal	2024
FRQS AI Summer School - Concordia University, Montreal	2023
INTER-REPAR Summer School - Université de Sherbrooke	2022
Drone Pilot Certificate (Basic Operations) – Transport Canada	2021
Programming of PLC S7-300 – Iran Technical & Vocational Training Organization	2014

Languages

Persian: Native | English: Fluent | French: B1 (TEFAQ)